Background Bayesian Meta-analysis

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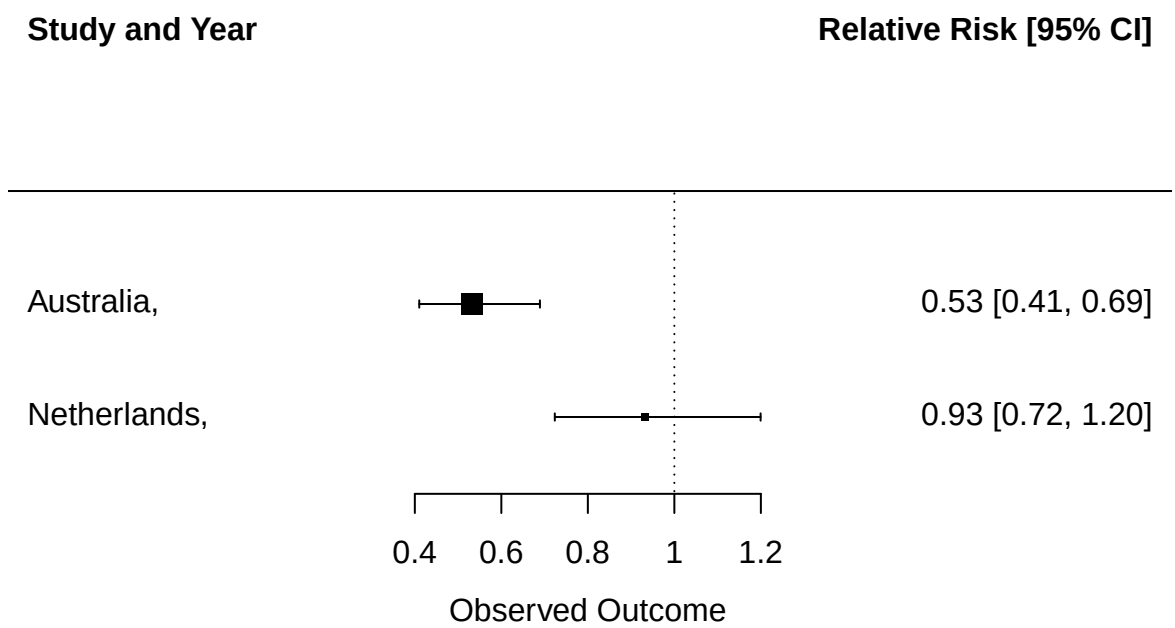
Colchicine in chronic CAD

A randomized, controlled, double-blind trial comparing colchicine to placebo, in patients with chronic coronary disease (LoDoCo2) was published in the NEJM. The study reports a total of 5522 patients underwent randomization; 2762 were assigned to the colchicine group and 2760 to the placebo group. The median duration of follow-up was 28.6 months. A primary end-point event occurred in 187 patients (6.8%) in the colchicine group and in 264 patients (9.6%) in the placebo group (incidence, 2.5 vs. 3.6 events per 100 person-years; hazard ratio, 0.69; 95% confidence interval [CI], 0.57 to 0.83; P<0.001).
This trial recruited patients from 2 countries and was analysed by ignoring any regional differences. However it may be of interest to analyse this data taking into account any between region variability. This seems appropriate if one wants to eventually estimate the likely result if the study was repeated in a third country.

The data

```
country <- data.frame(study = c("Australia", "Netherlands"))
country$d0 <- c(147,117)
country$n0 <- c(953, 1807)
country$d1 <- c(78,109)
country$n1 <- c(951,1807)

## @knitr relative_risk
country$p1 <- country$d1/country$n1
country$p0 <- country$d0/country$n0
country$rr <- country$p1/country$p0
country$lrr <- log(country$rr)
country$lse <- sqrt((1 - country$p1)/(country$p1 * country$n1) + (1 - country$p0)/(country$p0 * country
country$lower <- exp(country$lrr - qnorm(.975) * country$lse)
country$upper <- exp(country$lrr + qnorm(.975) * country$lse)

## @knitr forest_plot
library("metafor")
p <- forest.default(x = country$rr, ci.lb = country$lower, ci.ub = country$upper,
       slab = paste(country$study, country$year, sep = ", "), refline = 1)
text(min(p$xlim), .88 * max(p$ylim), "Study and Year", pos = 4, font = 2)
text(max(p$xlim), .88 * max(p$ylim), "Relative Risk [95% CI]", pos = 2, font = 2)
```



The relevant statistic for the meta-analysis is the relative risk ratio, p_{1j}/p_{0j}. We work with the log of the relative risk ratio, $y_j = log(p_{1j}) - log(p_{0j})$ with variance of each term calculated by the *delta* method as follows;

$$sigma_j^2 \approx \frac{1 - p_{1j}}{n_{1j}p_{1j}} + \frac{1 - p_{0j}}{n_{0j}p_{0j}}$$

Hierarchical model

The results from the separate RCTs can be modeled using a hierarchical model. We use the fact that the log of the relative risk is approximately normally distributed and assume that the random effects follow a normal distribution,

$$y_j \sim N(\theta_j, \sigma_j^2)$$

A hierarchical model applied to meta-analysis is typically referred to as a random-effect meta-analysis model in the medical literature. The parameters of the hierarchical model can be estimated in either a frequentist or a Bayesian framework. In a frequentist setup, point estimates (rather than probability distributions) for the parameters are estimated.

$$\theta_j \sim N(\mu, \tau)$$

There are, in general, three ways to estimate the random effects, θ_j
* no pooling, each study is separate $\tau = \infty$
* completing pooling all θ_j are from a common distribution, so essentially $\tau = 0$ - this is also known as fixed-effect meta-analysis model completely pools the relative risk estimates across studies. The overall mean is commonly estimated by taking an inverse-variance weighted average of studies.
* partial pooling, the hierarchical model is a compromise between the no-pooling and the complete-pooling estimates. The parameters of the hierarchical model can be estimated in either a frequentist or a Bayesian framework. In a frequentist setup, point estimates (rather than probability distributions) for the parameters are estimated.

```r
library(meta)
# no pooling
me.fe <- rma(country$lrr, country$lse^2, method = "FE")
cat("Completely pooled results with the point estimate followed by the 95%CI")
```

```
## Completely pooled results with the point estimate followed by the 95%CI
```

```r
c(exp(me.fe$b), exp(me.fe$ci.lb), exp(me.fe$ci.ub))
```

```
## [1] 0.7091 0.5917 0.8498
```

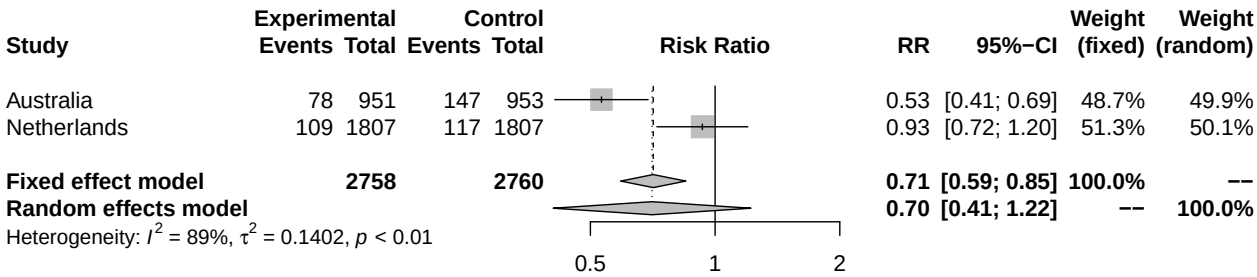
```r
#check to see that the point estimate is identical to taking a weighted average of the relative risks i
#exp(weighted.mean(country$lrr, 1/(country$lse^2)))

# frequentist random effects
me.re <- rma(country$lrr, country$lse^2)
cat("Hierarchical or random effects with the point estimate followed by the 95%CI")
```

```
## Hierarchical or random effects with the point estimate followed by the 95%CI
```

```r
c(exp(me.re$b), exp(me.re$ci.lb), exp(me.re$ci.ub))
```

```
## [1] 0.7044 0.4066 1.2204
```

```r
m1 <- meta::metabin(country$d1, country$n1, country$d0, country$n0, sm="RR", method ="I",
                    studlab=country$study, prediction=TRUE) #, comb.random =TRUE
meta::forest(m1)
```



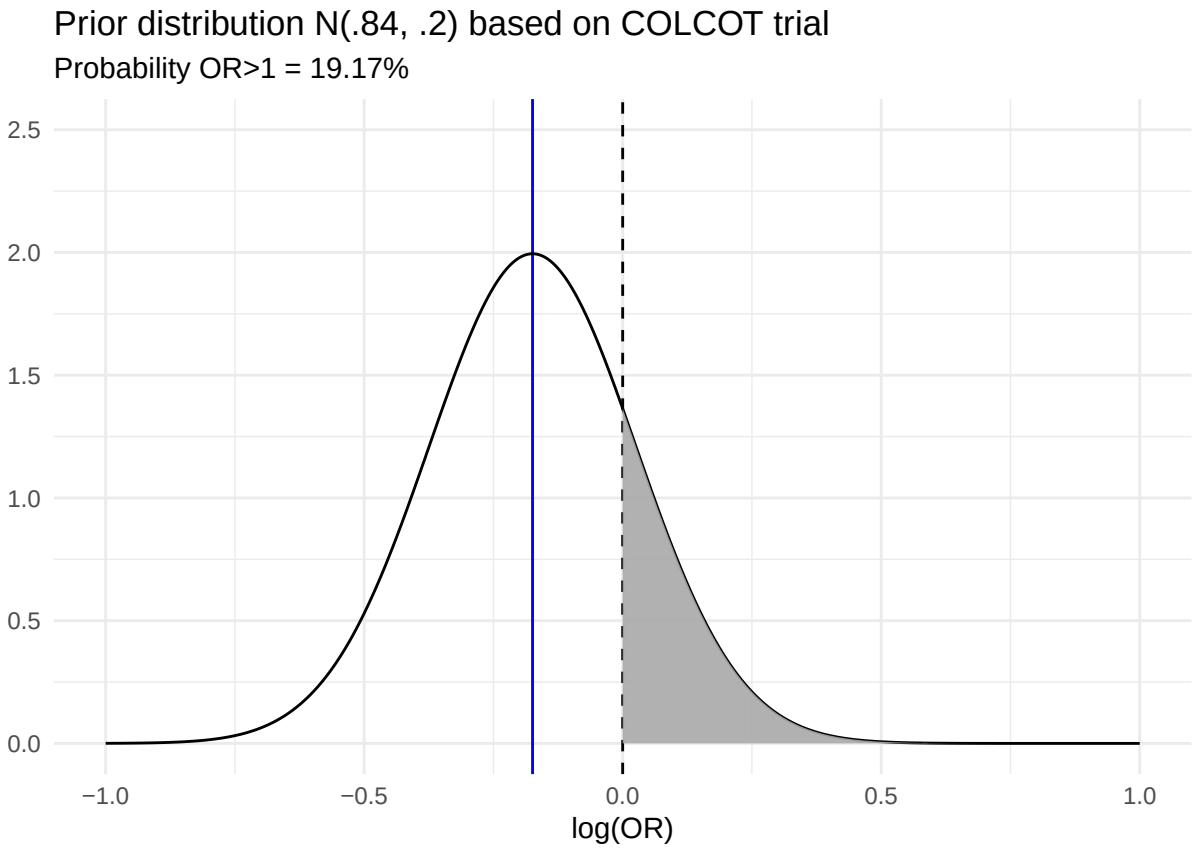
The bottom line appears that if one employs a model that captures the variability between different populations, it can be seen that while the point estimate suggests a benefit of colchicine in chronic CAD but the further studies are required to narrow our uncertainty. This is especially true since the outcome was a composite one largely driven by non-fatal outcomes

If one wants to adopt a bayesian approach we could incorporate previous RCT (COLCT) in a similar CAD population to refine this estimate while awaiting further studies.

Bayesian Approach

A similar randomized, double-blind trial comparing colchicine to placebo but involving patients recruited within 30 days after a myocardial infarction (COLCOT) was published in the NEJM and may provide a prior distribution for a Bayesian analysis.

This study reports a total of 4745 patients were enrolled; 2366 patients were assigned to the colchicine group, and 2379 to the placebo group. Patients were followed for a median of 22.6 months. The primary end point occurred in 5.5% of the patients in the colchicine group, as compared with 7.1% of those in the placebo group (hazard ratio, 0.77; 95% confidence interval [CI], 0.61 to 0.96; P = 0.02).

However, this study also ignores regional difference among the 12 participating countries. Therefore a reasonable prior might be centered at COLCOT point estimate but with increased variation is that there is have an approximately 20% probability of RR > 0, similar to the random effects result from LoDoCo2. This is attained with a distribution with$\mu \approx 0.8$ with $\sigma \approx .2$.

```r
my.func(my_sd=0.2,my_mean=log(0.84),prob_harm=TRUE,upper_limit=2.5)
```